Marcelo Lares | Senior Data Scientist | Machine Learning Engineer

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Senior Data Scientist and Machine Learning Engineer with experience developing production recommendation systems and leading client-facing AI initiatives. Specialized in ranking models, experimentation, feature engineering, statistical learning, and end-to-end model validation at LATAM scale. Former CONICET researcher and Associate Professor with a PhD in Astronomy, 30+ Q1 publications, and experience advising doctoral researchers.

Core Expertise

Machine Learning: PyTorch, Scikit-Learn, XGBoost, ranking, recommender systems, embeddings, feature engineering, computer vision

Production ML: ONNX, MLflow, feature stores, Fury pipelines, CUDA, A/B testing, Datadog, Looker

Data & AI: Python, SQL, R, NumPy, Pandas, SciPy, BigQuery, LangChain, RAG, prompt engineering

Engineering: Git, GitHub, Linux, APIs, Sphinx, AWS (S3/EC2), GCP (BigQuery/Cloud Storage)

Domains: E-commerce, retail, public revenue, tourism, agriculture, international trade

Professional Experience

Mercado Libre

Senior Data Scientist / Machine Learning Engineer, Argentina *2024–2026*

- Served as the primary data science owner for a production online ranking model powering recommendations for dozens of internal clients and millions of users across Argentina, Brazil, and Mexico.
- Developed and maintained the PyTorch ranking model, adding features, modifying embeddings, and integrating production feature-store data.
- Adapted Fury pipelines and optimized framework and CUDA configurations, tripling the amount of training data the model could process.
- Prepared ONNX artifacts, validated production configurations, and partnered with engineering to ensure reliable production deployments.
- Implemented MLflow experiment tracking and designed a pre-production validation protocol for safer, reproducible model releases.
- Developed statistical tools to quantify confidence in A/B-test differences and monitored clicks, purchases, GMV, and model behavior through Datadog and Looker.
- Delivered measurable lift in purchase conversion through controlled production experiments while partnering with engineering, product, and analytics teams.

IThreex Global

Lead Data Scientist, Argentina *2022–2024*

- Led an eight-person data science team delivering client-facing AI and analytics solutions across public revenue, retail, tourism, agriculture, and international trade.
- Designed and implemented the core Python library used by the Molibdeno AI platform to standardize reusable machine-learning workflows.
- Served as the principal technical contact for Kolektor, developing payment-behavior analyses and tax-revenue forecasts for large taxpayers in Córdoba.
- Delivered segmentation and predictive solutions that supported marketing and commercial strategies for Pueblo Nativo, Inverfin, and Agencia ProCórdoba.
- Developed computer-vision models for image-based cattle-weight estimation and defined the corresponding image-acquisition protocols.
- Built a LangChain-based RAG support service exposed through an API.
- Presented technical progress and results to clients and stakeholders, translating business requirements into data products.
- Mentored data scientists and supported them at specific projects

Academic and Research Background

Universidad Nacional de Córdoba

Associate Professor, Argentina *2023–present*

Head of the Data Science course in the Applied Mathematics department. Twenty years teaching statistics, machine learning, and scientific computing while advising graduate students. Now head of the Data Science course in the Applied Mathematics degree.

CONICET

Researcher, Argentina *2004–2023*

Applied Bayesian inference, statistical learning, numerical methods, and HPC to large astronomical datasets. Developed open-source scientific software and automated analysis pipelines, and collaborated in international multidisciplinary teams.

Education and Selected Achievements

Universidad Nacional de Córdoba

PhD in Astronomy, Argentina *2009*

Dissertation focused on statistical analysis of large-scale astronomical datasets.

Research: Author of 30+ peer-reviewed articles in international Q1 journals and multiple open-source scientific software projects; h-index 17.

Profiles: Google Scholar ORCID GitHub LinkedIn